

Fig. 5. BTL implementation of the ripple reduction circuit after the output filter of a Class-D amplifier.

amplifier, consisting of a MOS transistor ( $M_1$ ) which serves as a current source closing the feedback loop.

A key difference of this technique with respect to the other modulation techniques [3]–[7] is that the power is removed from the spectrum, instead of being spread out over neighboring frequencies. Another advantage compared to classic AB/D amplifiers is the high gain at  $f_{\text{PWM}}$ , which is made possible by the digital filter. A down-side of the single-ended implementation is high dissipation in  $M_1$ .  $M_1$  is biased in Class-A and always conducts a bias current while experiencing a  $V_{\text{DS}}$  of  $V_{\text{out}}$  of the amplifier.  $V_{\text{out}}$  is at half the supply and for a fixed bias current the dissipation of transistor  $M_1$  is proportional to  $V_{\text{out}}$ .

### B. Low Common-Mode Signaling

To reduce dissipation, the  $V_{\text{DS}}$  over  $M_1$  in Fig. 4 has to be reduced. By going to a BTL topology, a ternary modulation scheme [2] can be used that adds a degree of freedom to choose the CM level. Dissipation can be reduced by reducing the idle output CM voltage and thereby setting  $V_{\text{DS}}$  over the MOS transistors closer to ground. This BTL topology is shown in Fig. 5.

The top half of Fig. 6 shows the half bridge signals to have a low CM level. Achieving this lower CM level is done by subtracting an offset from the positive and negative half bridge signals. When the half bridge signals are high in amplitude, the subtracted offset will clip the signal at the lower bound  $V_{\text{D},0}$ . This  $V_{\text{D},0}$  is defined to keep the Class-A MOS transistors saturated. To maintain linearity, the part that was clipped off below  $V_{\text{D},0}$  is added to the other half bridge signal as shown in Fig. 6. The bottom half of Fig. 6 shows the DM and CM output signals. Note that the  $V_{\text{DM}}$  is the reconstructed sine and  $V_{\text{CM}}$  is at 5 V except when one of the half bridge signals is clipped below  $V_{\text{D},0}$ .

A low CM level reduces the idle power dissipation [14], but also reduces the performance of the amplifier. When one of the half bridges is clipped to  $V_{\text{D},0}$ , just one half bridge is driving the load. During this transition the loop filter has to process an instantaneous change in slope from the output of the amplifier.

In the top half of Fig. 7 the contribution of each half bridge for a certain output voltage is given. Around idle output, both the HBs are controlling the signal and once the output signal goes over approximately  $|V_{\text{DD}}|/4$  one of the HBs is clamped to  $V_{\text{D},0}$ . The bottom half of Fig. 7 shows the total

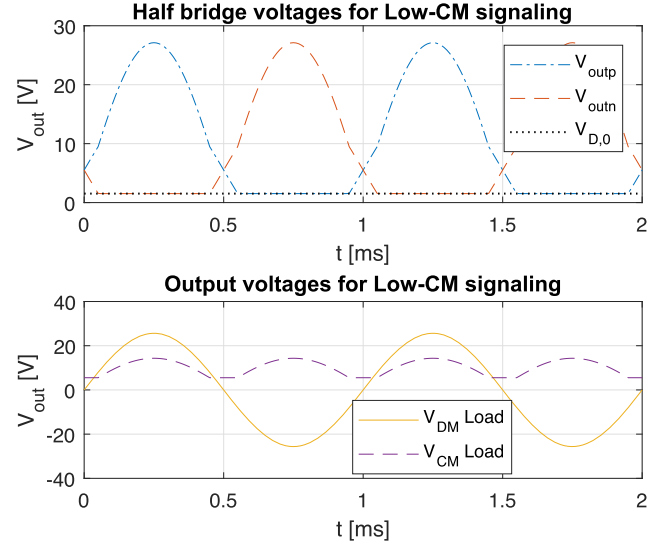


Fig. 6. Waveform illustrating the effect of low-CM signaling on half-bridge outputs (top) and CM and DM outputs (bottom).

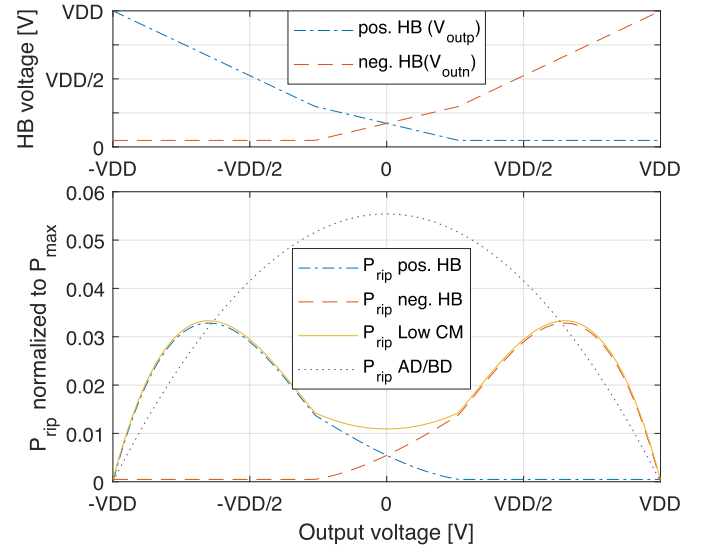


Fig. 7. Contribution of half bridges (HB) to the output voltage (top) and normalized power dissipation due to the ripple reduction.

dissipation as function of the instantaneous output voltage. The dissipation curve for active ripple reduction with conventional AD or BD modulation [2] is drawn to show the advantage in power dissipation of the low CM switching over normal AD or BD modulation. As audio has a high peak to average power ratio [15], most of the signal content resides in the valley of the graph and the ripple reduction power efficiency is up to 4 times better due to low-CM signaling.

### III. LOOP DESIGN

Technology scaling has led to smart power processes which allow to integrate low-power digital signal processing (DSP) and high-power amplifiers on the same die. This allows the design of Class-D amplifiers with a digital, high-order loop filter [16] and an analog-to-digital converter (ADC) to

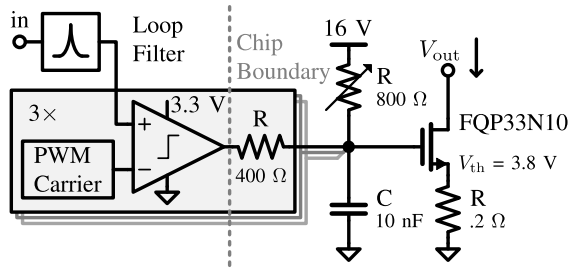


Fig. 8. Architecture of the DAC and Class-A driver.

provide feedback at the switching node [17] or after the output filter [18], [19]. Digital filters have an advantage over analog filters, being unaffected by process and temperature variation, allowing the use of higher order filters.

First, the Class-D amplifier is discussed. Secondly, all the blocks in the ripple reduction part of Fig. 5 will be discussed, as they define design constraints for the digital loop filter that is to be implemented. After that the loop design method is explained and finally compensation is added to ensure a stable system.

#### A. Digital Class-D Amplifier

To test the ripple reduction, a prototype was built around an AX5689 demo-board [20], as it features a high-performance audio amplifier, multiple ADCs and has a DSP core to implement a loop filter for the ripple reduction.

The core of the amplifier is a fully digital modulator and loop filter as seen in Fig. 5. A digital input signal is applied to the input. The feedback signal is digitized by the ADC and subtracted from the input, and the result is fed to the loop filter. The filter output is fed to the modulator, where the signal is compared to a digital staircase approximation of a triangle wave to obtain the PWM signals for the BTL power stage.

The ADC in the loop is a  $\Delta\Sigma$  ADC because of the high accuracy and low latency required in audio applications. Detailed design considerations of such a low-latency ADC (LLADC) and the amplifier are given in [19].

#### B. Class-A Driver

In order to realize the ripple reduction feedback loop, the digital feedback signal needs to be converted to the analog domain. The AX5689 is capable of generating eight PWM outputs, of which two are used for the BTL power stage. Hence the remaining six are distributed to have 3-phase PWM for each Class-A driver to perform digital-to-analog conversion. The schematic of this circuit is shown in Fig. 8.

Analog-to-digital conversion is done using one LLADC [19] per half bridge. The ripple reduction ADC output is fed to the loop filter, of which the output is fed to three PWM modulators. The modulators run at twice the loop filter sample rate, allowing 2 conversions per sample. The three carriers are shifted  $120^\circ$  in phase, and to reduce quantization noise, each carrier cycles through 6 different levels as shown in Fig. 9.

Summation of the PWM outputs is done using a resistive network that also provides level shifting and biasing to drive

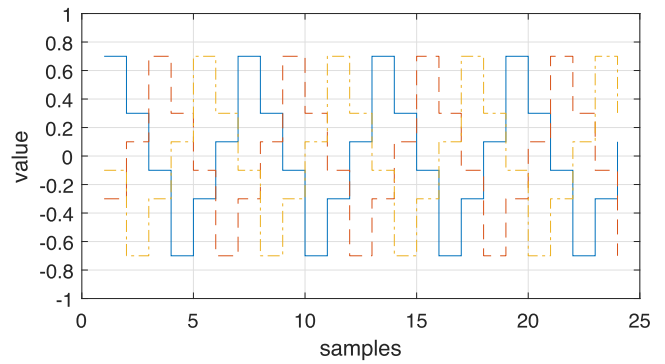


Fig. 9. 3-phase PWM carriers for D/A conversion.

the MOS transistor. The drain of the MOS transistor is connected after the LC-filter as in Fig. 5, closing the loop.

#### C. Loop Design Constraints

The ripple reduction loop consists of several blocks as can be seen in Fig. 5. The LC-filter impedance converts the MOS transistor drain current, which is injected at the output of the LC filter, to a voltage at the output of the LC filter. This impedance has a band-pass characteristic, having 2 poles and 1 zero. Spread in C affects the ripple reduction loop stability because it changes the gain for signals above the LC corner frequency.

The used ADC is also integrated in the AX5689 chip. It has a second order rising slope of 40 dB/dec between 60 kHz and 1 MHz due to the two zeroes at 60 kHz transitioning into  $-20$  dB/dec slope after 1 MHz due to the three poles at 1 MHz. The FIR DAC in the ADC renders the response highly insensitive to mismatch [19].

The RC filter used after the PWM DAC to reduce quantization provides an integrating response for signals at the PWM frequency due to the pole around 100 kHz. Spread in R and C will slightly alter the frequency of this pole and modify the integrator gain, this is however suppressed by the high overall loop gain. Finally the digital circuitry has processing delay, which gives a phase shift at frequencies in the vicinity of  $f_{s,loop}$ .

Summarizing, these blocks contribute the following poles and zeros,

- LC-filter (2 poles @30 kHz, 1 zero @DC)
- ADC (3 poles @1 MHz, 2 zeros @60 kHz)
- RC Integrator (1 pole @100 kHz)
- 3 unit delays (3 poles in the origin of the z-plane)

#### D. Filter Synthesis

The next objective is to design a loop filter that can be put in the ripple reduction feedback loop in Fig. 5. As was mentioned earlier, this filter requires high gain at  $f_{PWM}$  and low gain in the audio band. To design such a filter, a method common in  $\Delta\Sigma$  loop filter design [21] is used. With this method, a desired closed loop Noise Transfer Function (NTF) can be chosen and used to calculate the open loop transfer function  $H_{loop}(z)$ .

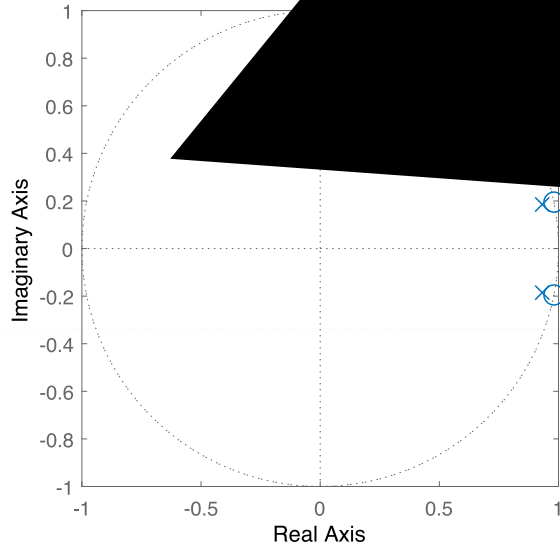


Fig. 10. Poles and zeros of the designed NTF.

This is done by rearranging the definition of the NTF in a feedback system as follows,

$$\text{NTF}(z) = \frac{1}{H_{\text{loop}}(z) + 1}, \quad (1)$$

$$H_{\text{loop}}(z) = \frac{1}{\text{NTF}(z)} - 1. \quad (2)$$

In this case, an NTF is chosen which leaves the audio intact and suppresses the PWM ripple. A way to design this is to place a notch at the PWM frequency. To make a notch, a complex zero pair is placed on the unit circle at an angle that is corresponding to the desired frequency. A complex pole pair is then placed inside the unit circle, at the same angle, close to the complex zeros in order to set the notch bandwidth. The angle in radians is calculated as follows,

$$\alpha = 2 \cdot \pi \frac{f_{\text{pwm}}}{f_{\text{s,loop}}}. \quad (3)$$

Now, using the exponential function the zero locations on the unit circle can be obtained as  $z_1 = e^{j\alpha}$  and  $z_2 = e^{-j\alpha}$ . The pole locations can be derived by multiplying the zeros with a factor  $r < 1$ , giving  $p_1 = r \cdot z_1$  and  $p_2 = r \cdot z_2$ . The closer  $r$  is to 1, the higher the quality factor and the sharper the notch will be.

To make a visualization of the poles and zeros, values have to be substituted in (3). A demo board with the 'AX5689' [20] is used to implement the filter and the audio amplifier. The default values in the controller are  $f_{\text{pwm}} = 768 \text{ kHz}$  and  $f_{\text{s,loop}} = 32 \cdot f_{\text{pwm}} \approx 25 \text{ MHz}$ , which is used for the loop filter and ADC. The pole-zero map is shown in Fig. 10. In the actual system  $r$  is set to 0.996 but for visualization purposes the value for  $r$  is 0.95.

To get from this closed loop NTF description to a loop filter that can be put in the feedback loop in Fig. 5, (2) is applied. The resulting loop filter has a resonator at  $f_{\text{pwm}} = 768 \text{ kHz}$ , because of the  $1/\text{NTF}$  operation in (2). The loop filter transfer function is shown in Fig. 11. From the phase plot it can be

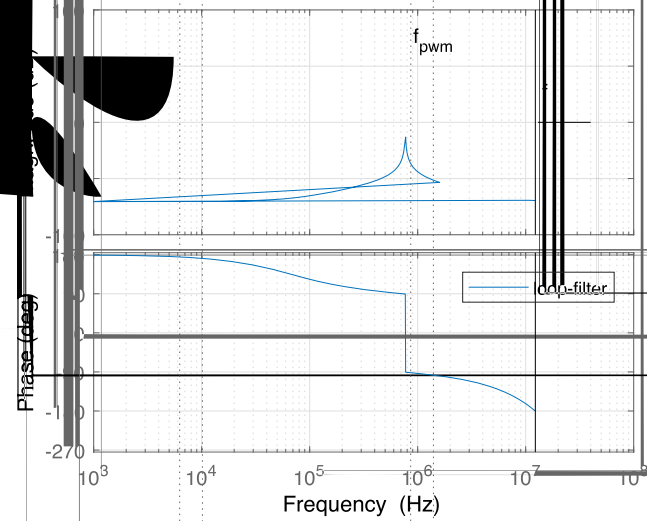


Fig. 11. Bode plot of the designed ideal loop filter.

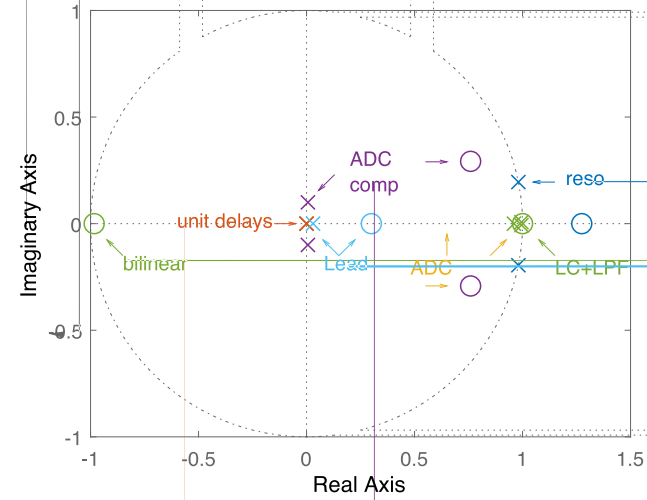


Fig. 12. Pole-zero map of the complete open loop transfer.

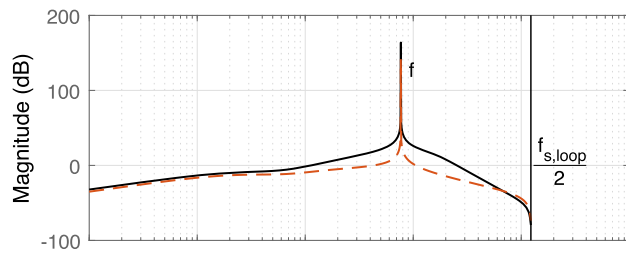
seen that around the resonant frequency, the phase is already  $\pm 90^\circ$ . Since the actual system consists of more elements than just the loop filter, maintaining stability becomes a challenge.

### E. Loop Compensation

Several blocks influence the stability of the ripple reduction loop. Hence, compensation has to be added to make the loop robust. All poles and zeros have been plotted in Fig. 12.

In order to compensate the ADC poles, a second order lead compensator is added 'ADC comp' in Fig. 12. The complex pole pair of the 3 ADC poles is canceled by placing a complex zero pair on top of them. To maintain causality, two poles are added near  $z = 0$ . Ideally these could be unit delays.

The additional phase shift that the unit delays create is compensated by moving the zero of the resonator 'reso' more to the right, thereby reducing the phase lag near the resonator. This zero lies outside the unit circle, which is equivalent to a right half plane zero in the  $s$ -domain.



Frequency (Hz)







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